Behavior Codebook V5

Background

Participants collaborate on the NASA Moon Survival Task, a group decision-making exercise in which they rank survival items by priority. They discuss five items selected from the original NASA Moon Survival list and arrange the ranking using a shared touchscreen.

Code Scheme

Behavior Label	Definition
Problem	Naming a problem **OR** describing a problem. Naming a problem is pointing out that a problem exists, without elaboration. Describing a problem is explaining or giving examples of what the problem is.
Solution	Identifying a solution **OR** describing a solution. Identifying a solution is introducing a new idea, option, or modification of an existing option that addresses the content of the problem itself. Describing a solution is illustrating or explaining what a solution is.
Concern with a solution	Pointing out a problem or a concern of a proposed solution, state the reasons for the opposition, or explain why the solution won't work. Supporting visual cues could be skeptical facial expression (e.g. frown) or posture (e.g. shake the head) while speaking.
Arguing for a solution	Giving reasons to support a proposed solution, primarily including its advantages, evidence, vision, expected result,etc. Supporting visual cues could be a sympathetic facial expression and posture (e.g. nod) while speaking.
Managing procedures	The person directs the group's working process rather than contributing to the substance of the problem or its solutions. Such talk keeps the discussion focused, structured, and moving forward (e.g., procedural questions, procedural suggestions, procedural clarifying, prioritizing, time management, task distribution and summarizing), but does not itself add problem or solution content, nor commit the group to a final choice.
Others	If none of the Verbal labels above applies.

Behavior Label	Definition
Making decision	Registering a decision by the physical action of touching the shared touchscreen and rearranging options on it. The key visual cue is a deliberate drag/slide gesture. Supporting verbal cues could be announcing and describing what the decision is. Making decision does NOT include brief or habitual screen contact, e.g. tapping, drumming fingers, resting a hand on or near the screen, or pointing at the screen while talking.
Laugh	The person laughs or smiles, either as a vocal laughter or as a facial expression alone. A laugh isn't a momentary action; it encompasses the entire process, from the initial change in facial expression to the moment the laugh fades. Any visible smiling or laughing expression counts.

Unitizing

We analyze the expression unit, which constitutes the smallest meaningful segment of events that can be coded into the behaviors of the code scheme. In our study, this is usually a **dialogue turn or a continuous behavior**.

Short, fragmentary utterances that do not express a complete meaning on their own do **not** form a unit and receive no label: for example, ”Ok”, ”agree”, ”yeah”, or an utterance that breaks off mid-thought, such as ”I think we could—”. When such words lead into a full sentence, they are merged into that sentence. In contrast, a short sentence that expresses a clear, self-contained meaning for the task **does** form a unit and is labeled; for example, short sentences that move the group process forward, such as ”I agree with it” or ”How about you?”, are coded as Managing procedures.

A new unit is parsed when one of the following occurs:

- The speaker changes (turn-taking).
- There is a long pause longer than **6** seconds.
- The speaker is interrupted mid-sentence.
- Within a conversational turn, the speaker voices several statements where each expresses a complete thought (e.g., first describing a problem and then asking a question).

Every speech unit receives exactly one label. A speech unit that matches none of the five Verbal codes is labeled Others. Nonverbal behaviors are coded only when they occur. Behaviors of the same type cannot overlap, but behaviors of different types can overlap.

Distinguishing Easily Confused Codes

- **Problem vs. Concern with a solution**: If the problem is raised about a specific proposed solution, code it as Concern with a solution; if it is independent of any proposed solution, code it as Problem. A concern usually carries an opposing stance toward the proposed solution, while a problem is stance-neutral.
- **Solution vs. Arguing for a solution**: Proposing a solution, explaining how its mechanism works, or giving examples of the mechanism counts as Solution. Arguing for a solution

does not explain the mechanism itself; it explains why the mechanism would work, such as its advantages and expected results, and usually carries a supportive stance toward the proposed solution.

Examples

1. Speech:

”person”: ”Person A”,

”start_time”: ”00:06:28:02”,

”end_time”: ”00:06:37:24”,

”behavior”: ”Concern with a solution”,

”person”: ”Person B”,

”start_time”: ”00:04:51:38”,

”end_time”: ”00:04:57:32”,

”behavior”: ”Arguing for a solution”,

”person”: ”Person A”,

”start_time”: ”00:04:28:15”,

”end_time”: ”00:04:37:25”,

”behavior”: ”Problem”,

”person”: ”Person A”,

”start_time”: ”00:05:26:10”,

”end_time”: ”00:05:32:03”,

”behavior”: ”Solution”,

”person”: ”Person B”,

”start_time”: ”00:06:40:17”,

”end_time”: ”00:06:46:20”,

”behavior”: ”Managing procedures”,

2. Vision:

”person”: ”Person A”,

”start_time”: ”00:05:59.460”,

”end_time”: ”00:06:03.910”,

”behavior”: ”Laugh”,

”person”: ”Person A”,

”start_time”: ”00:06:12.000”,

”end_time”: ”00:06:19.070”,

”behavior”: ”Making decision”,